

LLMGrid Codebase Navigation Guide

Supplementary Material for “Telepathy vs. Talk”

1 Reproducibility: Data and Code Locations

This guide provides exact locations for all experimental data and analysis scripts. The two experimental paradigms live on separate Git branches with distinct implementations.

1.1 Communication Branch Experiments (Git Branch: main)

Implementation: The Communication experiments use the `main` branch codebase, which implements:

- `goal_sensor` field in agent observations (bearing + strength hints)
- ASCII map interface (`world_map_ascii`)
- Explicit message passing (INTENT, REQUEST, CHAT protocols)
- 117-line dynamic prompt with communication rules

Git Reference: Branch `main`, commit `03c4763`

Experiment Directory: `experiments/cross_seed_baseline_20251112T143355Z/`

Data Runs (N=45):

- Seeds 13–17, 3 strategies (none, structured, freeform), 3 replicas each
- Episode data: `runs/*/results/episode.json`
- Transcripts: `runs/*/results/transcript.jsonl` (includes `goal_sensor` observations)
- Per-run metrics: `runs/*/results/metrics.json`

Aggregated Data:

- `run_inventory.json` - Complete run metadata with message counts
- `aggregate_stats.json` - Computed statistics across all 45 runs

Direct Link: https://github.com/Bryan-Nsoh/agent-talk-2/tree/main/experiments/cross_seed_baseline_20251112T143355Z

1.2 Map-Sharing Branch Experiments (Git Branch: map-share)

Implementation: The Map-Sharing experiments use the `map-share` branch codebase, which implements:

- No `goal_sensor` field (agents explore blindly)
- Full grid JSON interface (`grid.rows` 2D array)
- No explicit messaging (communication disabled)
- Map synchronization modes: `none`, `radio_sync`, `global`
- 76-line static prompt without communication rules

Git Reference: Branch `map-share`, commit `117ebaa`

Experiment Directory: `experiments/mapshare_long_corridor_20251119T202017Z/`

Data Runs (N=45):

- `none/runs/*` - Baseline (no map sharing, N=15)
- `radio_sync/runs/*` - Radio-range map merging (N=15)
- `global/runs/*` - Global map sharing (N=15)
- All runs include: `transcript.jsonl`, `episode.stream.jsonl`, `config.yaml`

Direct Link: https://github.com/Bryan-Nsoh/agent-talk-2/tree/map-share/experiments/mapshare_long_corridor_20251119T202017Z

1.3 Analysis Scripts (Branch: main)

All statistical analysis scripts live on the `main` branch and are branch-agnostic (work with data from both experiments):

- `analysis/extract_raw_data.py` - Branch-agnostic data extraction
 - Parses `episode.json` (uses `finished` list for map-share, `status` field for communication)
 - Parses `transcript.jsonl` for collision counts (handles both `last_move_outcome` and `last_result.kind`)
 - Outputs: `analysis/raw_data.json` (N=90: 45 communication + 45 map-share)
- `analysis/statistical_tests.py` - Mann-Whitney U tests
 - One-tailed tests (`alternative='greater'`) for Map-Sharing
 - Two-tailed tests (`alternative='two-sided'`) for Communication
 - Outputs: `analysis/statistical_results.json` with all p-values

Plot Generation:

- Communication plots (branch `main`):
 - Source: `experiments/cross_seed_baseline_20251112T143355Z/plots/`

- Script: `experiments/.../generate_plots_45runs.py`
- Map-Sharing plots (branch main):
 - Source: `analysis/mapshare/plots/`
 - Script: `analysis/mapshare/generate_plots.py`

1.4 Quick Reference: Direct GitHub Links

Communication (Branch main):

- Experiment data: https://github.com/Bryan-Nsoh/agent-talk-2/tree/main/experiments/cross_seed_baseline_20251112T143355Z
- README with full results: https://github.com/Bryan-Nsoh/agent-talk-2/blob/main/experiments/cross_seed_baseline_20251112T143355Z/README.md

Map-Sharing (Branch map-share):

- Experiment data: https://github.com/Bryan-Nsoh/agent-talk-2/tree/map-share/experiments/mapshare_long_corridor_20251119T202017Z
- README with full results: https://github.com/Bryan-Nsoh/agent-talk-2/blob/map-share/experiments/mapshare_long_corridor_20251119T202017Z/README.md

Analysis Scripts (Branch main):

- Statistical tests: https://github.com/Bryan-Nsoh/agent-talk-2/blob/main/analysis/statistical_tests.py
- Data extraction: https://github.com/Bryan-Nsoh/agent-talk-2/blob/main/analysis/extract_raw_data.py
- Raw data (N=90): https://github.com/Bryan-Nsoh/agent-talk-2/blob/main/analysis/raw_data.json
- Test results: https://github.com/Bryan-Nsoh/agent-talk-2/blob/main/analysis/statistical_results.json